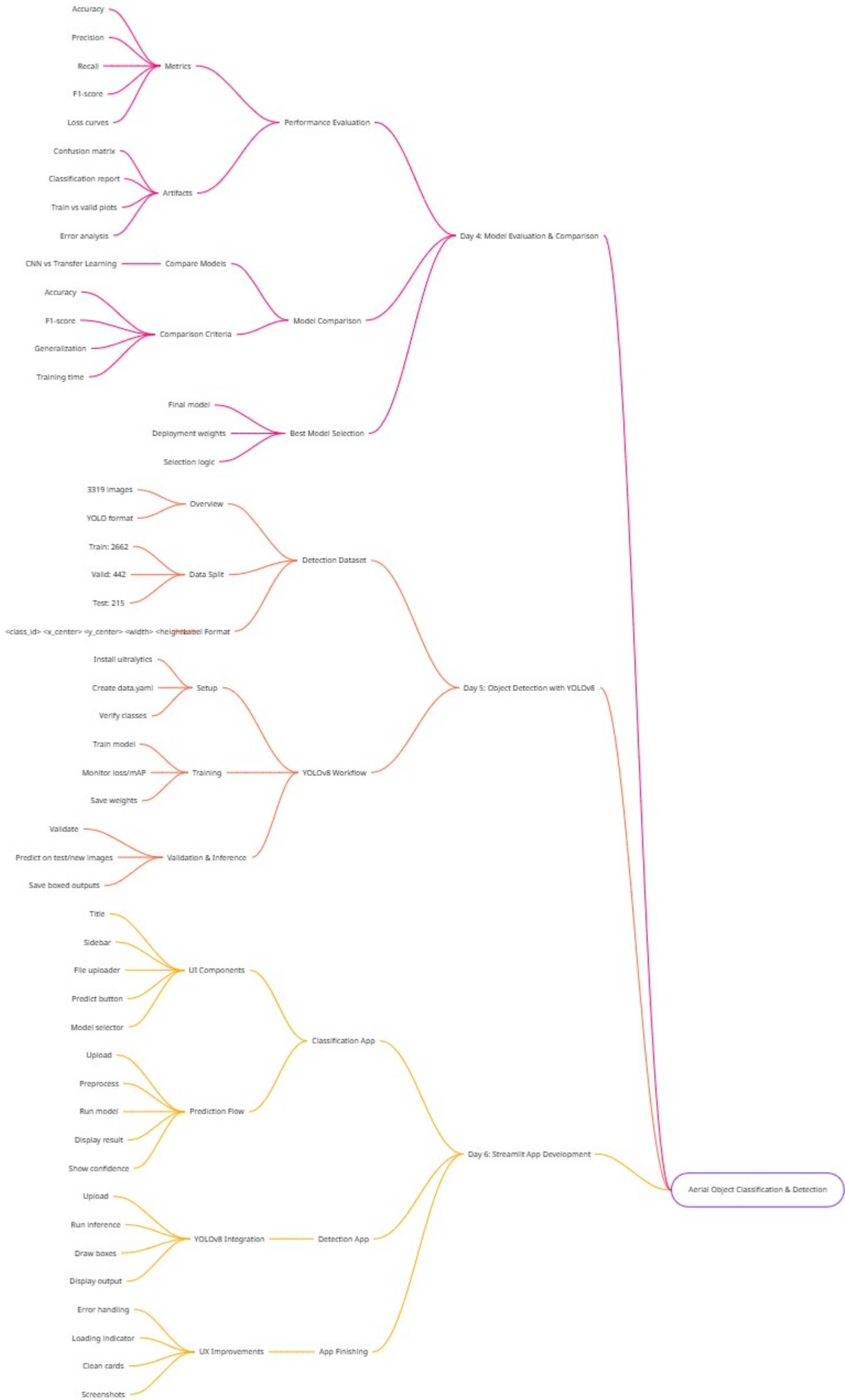
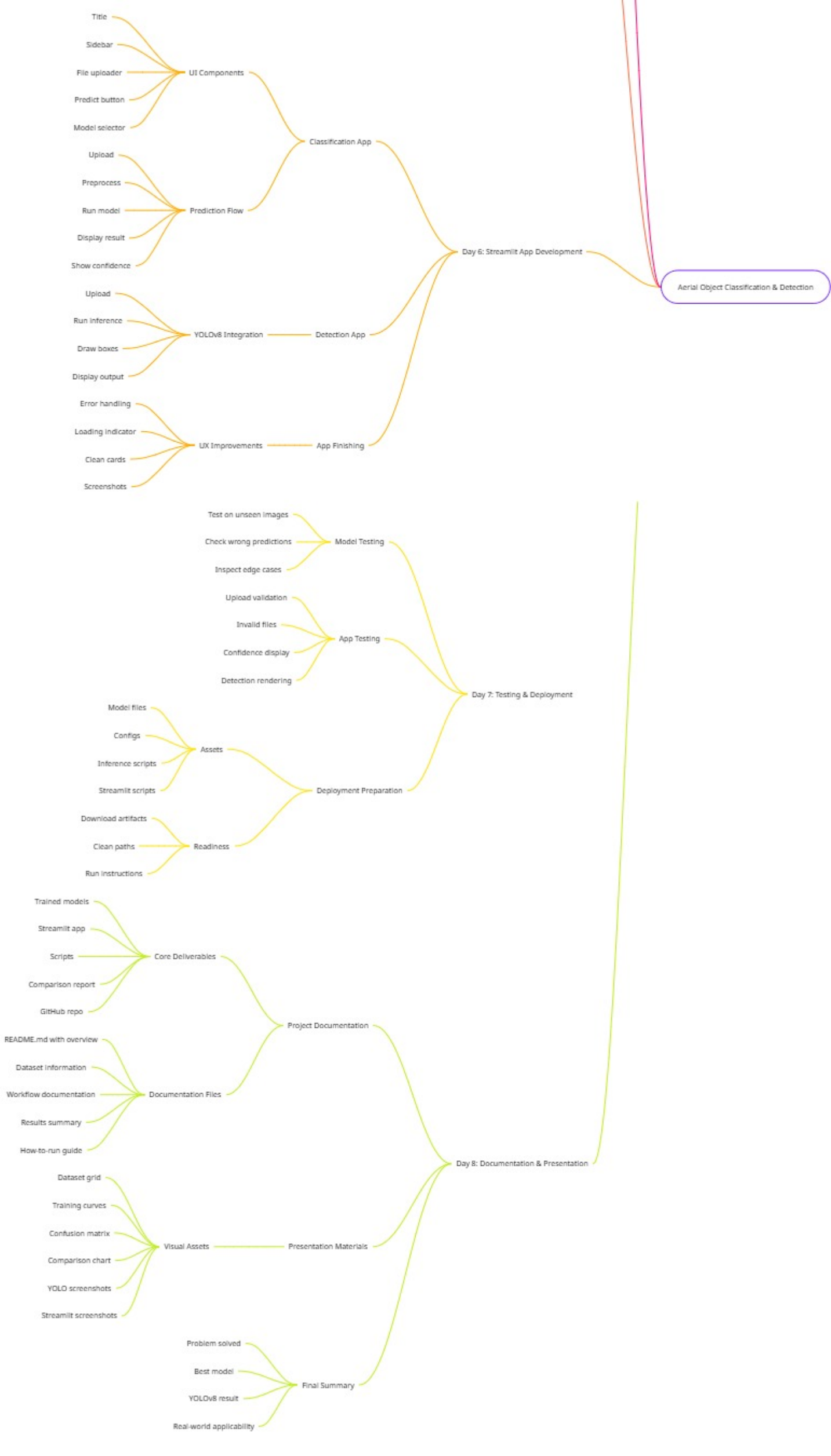
- Aerial Object Classification & Detection
  - Day 1: Project Planning & Setup
    - Define Project Scope & Requirements
      - Problem Statement
        - Classify Bird vs Drone
        - YOLOv8 detection
        - Streamlit deployment
      - Skills/Tools
        - Python
        - Deep Learning
        - Computer Vision
        - TensorFlow/PyTorch
        - Transfer Learning
        - YOLOv8
        - Streamlit
      - Real-World Use Cases
        - Wildlife protection
        - Security surveillance
        - Airport bird-strike prevention
        - Environmental research
    - Environment Setup
      - Colab Workspace
        - Mount Drive
        - GPU runtime
      - Project Folder Structure
        - data/
        - notebooks/
        - src/
        - configs/
        - models/
        - figures/
        - reports/
        - app/
      - Config Initialization
        - paths.yaml
        - classification_config.yaml
        - detection_config.yaml
        - streamlit_config.yaml
  - Day 2: Data Collection & Preprocessing
    - Classification Dataset
      - Dataset Inspection
        - RGB .jpg
        - Bird/Drone classes
      - Data Split
        - Train: Bird 1414, Drone 1248
        - Valid: Bird 217, Drone 225
        - Test: Bird 121, Drone 94
      - Initial Analysis
        - Folder structure
        - Class imbalance
        - Sample visualization
    - Image Preprocessing
      - Standard Preprocessing
        - Resize 224×224
        - Normalize [0,1]
        - Label conversion
      - Data Pipeline Setup
        - Train/Valid/Test loaders
        - Batch preview
    - Data Augmentation
      - Techniques
        - Rotation
        - Flip
        - Zoom
        - Brightness
        - Cropping
      - Validation
        - Display samples
        - Label consistency
  - Day 3: Classification Model Development
    - Custom CNN Model
      - Architecture
        - Conv blocks
        - Max pooling
        - Batch norm
        - Dropout
        - Dense head
      - Training Setup
        - Loss function
        - Optimizer
        - Batch size
        - Epochs
        - Early stopping
      - Training Execution
        - Train/validate
        - Save weights
        - Save history
    - Transfer Learning Models
      - Candidates
        - MobileNet
        - ResNet50
        - EfficientNetB0
      - Workflow
        - Load pretrained
        - Freeze base
        - Custom head
        - Train top layers
        - Fine-tune
      - Outputs
        - Best model
        - Validation accuracy
        - Training curves


Aerial Object Classification & Detection
Day 4: Model Evaluation & Comparison
Performance Evaluation
Metrics
Accuracy
Precision
Recall
F1-score
Loss curves
Artifacts
Confusion matrix
Classification report
Train vs valid plots
Error analysis
Model Comparison
Compare Models
CNN vs Transfer Learning
Comparison Criteria
Accuracy
F1-score
Generalization
Training time
Best Model Selection
Final model
Deployment weights
Selection logic
Day 5: Object Detection with YOLOv8
Detection Dataset
Overview
3319 images
YOLO format
Data Split
Train: 2662
Valid: 442
Test: 215
<class_id> <x_center> <y_center> <width> <height> Label Format
YOLOv8 Workflow
Setup
Install ultralytics
Create data.yaml
Verify classes
Training
Train model
Monitor loss/mAP
Save weights
Validation & Inference
Validate
Predict on test/new images
Save boxed outputs
Day 6: Streamlit App Development
Classification App
UI Components
Title
Sidebar
File uploader
Predict button
Model selector
Prediction Flow
Upload
Preprocess
Run model
Display result
Show confidence
Detection App
YOLOv8 Integration
Upload
Run inference
Draw boxes
Display output
App Finishing
UX Improvements
Error handling
Loading indicator
Clean cards
Screenshots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Day 7: Testing & Deployment
Model Testing
Test on unseen images
Check wrong predictions
Inspect edge cases
App Testing
Upload validation
Invalid files
Confidence display
Detection rendering
Deployment Preparation
Assets
Model files
Configs
Inference scripts
Streamlit scripts
Readiness
Download artifacts
Clean paths
Run instructions
Day 8: Documentation & Presentation
Project Documentation
Core Deliverables
Trained models
Streamlit app
Scripts
Comparison report
GitHub repo
Documentation Files
README.md with overview
Dataset information
Workflow documentation
Results summary
How-to-run guide
Presentation Materials
Visual Assets
Dataset grid
Training curves
Confusion matrix
Comparison chart
YOLO screenshots
Streamlit screenshots
Final Summary
Problem solved
Best model
YOLOv8 result
Real-world applicability